Tiago Aleixo

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Education

University of Porto — Faculty of Sciences Expected Jun. 2027
MSc Network & Information Systems Engineering *Porto, Portugal*

- Current average: 14/20.
- Relevant coursework: Embedded Systems, Network Security, Distributed Systems, Cloud Systems Administration, Systems Administration, Advanced Computer Networks.

University of Porto — Faculty of Engineering Jun. 2025
BSc Informatics & Computing Engineering *Porto, Portugal*

- Final average: 13/20.
- Relevant coursework: Computer Architecture, Operating Systems, Computer Networks, Compilers, Parallel & Distributed Computing, Computer Security.

Technical Skills

Languages: C, C++, Python, Java, JavaScript, Bash, SQL
Systems: Linux, QEMU/KVM, Proxmox VE, Docker, Kubernetes, Terraform
Embedded: Zephyr RTOS, Nordic nRF SDK, ESP32/ESP8266, BLE, SPI, I2C, UART, JTAG/SWD
Networking: TCP/IP, VLANs, WireGuard, OPNsense, Suricata IDS/IPS
Graphics: WebGL, GLSL, 3D rendering, shaders

Experience

Google Jul. 2026 – Sep. 2026
Software / Operations Engineering Intern *Groningen, Netherlands*

- Engineered an extensible workflow orchestration and milestone tracking platform, unifying 60+ siloed data sources into an automated, event-driven data ingestion pipeline.
- Built a two-way synchronization layer with scoped mutex locking, providing transactional consistency across distributed endpoints.
- Implemented a fault-tolerant state checkpointing engine that tracked execution budgets, serialized runtime state, and self-scheduled continuation jobs to prevent data loss.
- Developed a real-time diagnostics dashboard for hardware triage outcomes and platform error logs, automating root-cause analysis and reducing manual triage latency.
- Delivered 5,000+ lines of production code and 4,000+ lines of unit tests; anonymized telemetry measured 99.99% production reliability, reclaiming 70+ engineering hours/month across 12 EMEA operational sites.

Fraunhofer AICOS Jul. 2025 – Jul. 2026
Embedded Systems Researcher *Porto, Portugal*

- Architected and implemented an indoor real-time location system from scratch on Zephyr RTOS using a Nordic nRF52-based SoC, supporting concurrent multi-tag tracking over BLE.
- Developed multilateration and signal-filtering algorithms achieving sub-2m positioning accuracy in high-interference real-world environments.
- Deployed the complete hardware/software stack in an active hospital environment, tracking 20+ medical assets in real time.
- Built the software platform for real-time data ingestion, REST/WebSocket APIs, and interactive visualization.

Schmitt + Sohn Elevadores Feb. 2025 – May 2025
Software Engineering Intern *Porto, Portugal*

- Developed backend services with Java Spring Boot and PostgreSQL, implementing REST APIs and optimizing database queries.

Selected Projects

NES Emulator
C, MINIX 3, 6502, PPU, UART

- Built an NES emulator in C on MINIX 3, implementing the 6502 CPU instruction set, memory addressing modes, and cycle-accurate execution.
- Implemented the NES PPU with framebuffer-based 256×240 rendering, including tile/sprite rendering, scrolling, and palette management.
- Validated CPU emulation against the NESTest ROM and debugged CPU/PPU timing and graphics rendering issues.
- Developed custom MINIX 3 device drivers and a UART interface for physical NES controller input.

Java Compiler
Java, ANTLR, OLLIR, Jasmin
- Built an end-to-end compiler for a Java subset, implementing lexical analysis, parsing, AST construction, symbol tables, semantic analysis, and type checking.
- Developed an OLLIR intermediate-representation pipeline with constant folding, constant propagation, and register allocation.
- Implemented a Jasmin backend generating JVM bytecode, completing the source-to-bytecode compilation pipeline.

Production Cluster & Perimeter Network Infrastructure
Proxmox VE, QEMU/KVM, TrueNAS, OPNsense
- Designed and operated virtualization and network infrastructure supporting 50+ active branch members, including Linux VMs, storage, VLANs, VPNs, and firewalling.
- Administered QEMU/KVM virtualization through Proxmox VE and deployed Suricata IDS/IPS and WireGuard-based network segmentation.

Leadership

IEEE UP Student Branch — Computer Society Mar. 2025 – Present
Computer Society Chair *Porto, Portugal*
- Led infrastructure initiatives with IEEE Portugal IT Systems, including deployment of a Proxmox/TrueNAS cluster supporting 50+ members.
- Organized technical workshops and hands-on sessions for 40+ students.